Study on the Dry Salt Playa

Sedimentary Stratification, Brine Geometries & Solar Evaporation

In the dry basin between the Sentinel Mesa and the sanctuary threshold lies the ancient playa. During winter flash floods, torrential waters descend from the Red Ridge, dissolving mineral salts from the sandstone bedrock. When the ferocious summer winds (Shamal) sweep across the plains, the moisture evaporates at an alarming rate, leaving polygonal crusts of interlocking halite prisms.

Fig. I - Stratigraphy of the Basin Floor

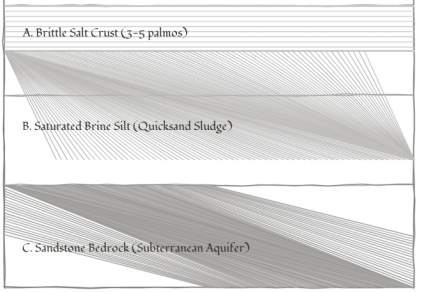
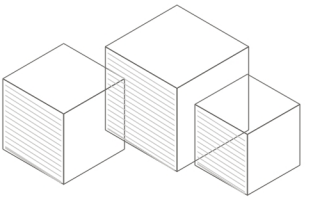


Fig. II - Isometric Halite Cleavage Prisms



Geometrical Observation on Salt Cracking:
When desiccation reaches its maximum, the crust shrinks towards centers of tension, forming hexagonal networks identical to the basalt columns of alpine crags and the drying of wet clay. The angle of intersection between adjacent fissures tends perpetually toward 120 degrees. Under intense solar rays, salt tears precipitate into rhomboid crystals of extreme hardness.

Nota dell'Architetto (Field Observation):

> «The water that flows carries the memory of the heights; but the salt that remains is the crystal skeleton of the earth, holding the light long after the storm has turned to dust. Let the seeker who traverses the playa walk with measured footsteps, for weight without balance yields to the deep beneath.»